BUGATTI HYPERCAR RFP PROPOSAL: ZF Lifetec (Passive Safety Systems)

Location: Alfdorf, Germany | Tier-1 Passive Safety Leader supplying Porsche Taycan Turbo GT, Audi R8, Lamborghini Revuelto a

RFP RESPONSE: BUGATTI-W16-V16-HYPER-RESTRAINT-2026
SUBMITTED BY: ZF Lifetec GmbH (voormals ZF Passive Safety Systems, Alfdorf, Germany)
OEM SUPERCAR PEDIGREE: Supplier of ACR8 Active Control Retractors and luxury illuminated/heated buckles for Lamborghini, Porsche, and BMW M.

1. CRASH PHYSICS, PYROTECHNICS & GLOBAL HOMOLOGATION
- Homologation Coverage: Full UN/ECE R16, FMVSS 209/210, GB 14166, and TRIAS road certification. (FIA 8853 track harness offered via modular drop-in kit).
- Pyrotechnic Retractor & Buckle: Next-gen ACR8.S combines reversible electro-mechanical tensioning with a 5.4 millisecond dual-stage pyrotechnic squib.
- Load Limiter Architecture: Adaptive 3-stage torsion bar (4.5 kN -> 3.4 kN -> 2.7 kN) with high-speed webbing payout damping.
- Pre-Crash Integration: Industry-leading ACR8 active haptic pulsing (warns driver via high-frequency belt vibrations) and 220 N pre-crash tensioning during 0.8G braking or drift/oversteer intervention.

2. HYPERCAR LIGHTWEIGHTING & EXOTIC MATERIALS
- Total Installed Vehicle Mass (2 Seats Complete): 2,190 grams per car (1,095 grams per seat).
- Hardware Metallurgy: Magnesium-aluminum die-cast retractor housing paired with CNC Ti-6Al-4V titanium load anchors.
- Webbing Spec: 1.10 mm multi-filament high-tenacity polyester/aramid edge webbing, 27.6 kN breaking strength, 5.1% elongation.

3. BUGATTI BESPOKE CRAFTSMANSHIP, LUXURY & ACOUSTICS/NVH
- Bespoke Customization: Dedicated Small-Series Luxury Cell in Alfdorf supports 50+ Pantone colors (MOQ 1 vehicle) plus optional fiber-optic ambient edge illumination.
- UV & Abrasion Resistance: Delta-E = 0.95 after 1,500 hours Xenon-Arc. 65,000 cycle abrasion rating.
- Tactile & Acoustic Engineering: Billet aluminum buckle housing wrapped in Molsheim-matched leather, featuring optional heated buckle tongue (warms to 36C in winter). Buckle click engineered to 55.2 dB.
- NVH Performance: Zero rattle (0 dB across 10-1,500 Hz) with magnetic tongue damper and active motorized presenter integration.

4. MANUFACTURING QUALITY & ZERO-DEFECT TRACEABILITY
- Certifications: IATF 16949:2016, ISO 14001, VDA 6.3 Audit Score 96% (Rating A).
- Historical Defect Rate: 0.6 PPM.
- Inspection Protocol: 100% 3D CT X-ray scan of pyrotechnics and gears, automated laser serializing, 35-year digital twin retention.

5. COMMERCIAL ECONOMICS & LOW-VOLUME NRE STRUCTURE
- Upfront NRE & Homologation Sled Testing: EUR 920,000 total (slightly above the EUR 850,000 target due to ACR8 active software calibration).
- Unit BOM Price (Per 2-Seat Vehicle Shipset): EUR 2,720.
- Bespoke Sur Mesure Color Surcharge: EUR 440 per custom color vehicle.

6. SUPPLY CHAIN RESILIENCE & MOLSHEIM ATELIER LOGISTICS
- Delivery to Molsheim: 5.0-week lead time; Alfdorf facility is only 195 km across the Rhine from Molsheim, enabling rapid 3-hour direct courier delivery in custom flight cases.
- Sub-Tier Sourcing: 100% German/European sub-tiers. 15-year structural warranty and 35-year classic spares program.